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Klasifikasi Risiko Kehamilan Menggunakan Random Forest Berdasarkan Rule-Based Klinis Maternal

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Abstract

High-risk pregnancy remains a significant challenge in maternal healthcare, requiring early and accurate identification during Antenatal Care (ANC) visits to prevent complications. This study aims to develop a machine learning-based pregnancy risk classification system using the Random Forest algorithm with maternal clinical rule-based labeling. The dataset consisted of 10,000 maternal clinical records obtained from ANC examinations, including features such as maternal age, blood pressure, hemoglobin level, body mass index, and disease history. Data preprocessing, feature engineering, random oversampling, and hyperparameter tuning using GridSearchCV with 5-fold cross-validation were applied to improve classification performance. The proposed model classified pregnancy risk into three categories: Normal, Need Treatment, and Need Referral. Experimental results showed that the optimized Random Forest model achieved a superior classification accuracy of 98.50%, effectively outperforming the Logistic Regression baseline model which yielded an accuracy of 95.65%. Feature importance analysis indicated that blood pressure, hemoglobin level, and clinical risk indicators contributed significantly to the classification process. The developed model demonstrates the potential of machine learning to support early pregnancy risk screening and assist healthcare workers in maternal risk assessment..

Keywords: Antenatal Care; Maternal Health; Pregnancy Risk; Random Forest; Rule-Based Labeling

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1. Introduction

Maternal mortality rates (MMR) and various clinical complications during pregnancy and childbirth remain critical challenges within the global healthcare system, particularly in primary care facilities. Continuous and regular monitoring through Antenatal Care (ANC) visits is the primary pillar recommended to achieve early screening of pathological risk factors in pregnant women. However, the maternal risk stratification process in the field is frequently hindered by the subjectivity of manual assessments and the overwhelming volume of clinical data that healthcare workers must evaluate rapidly. This limitation in decision support instruments exacerbates the likelihood of delays in administering early interventions or critical medical referrals crucial for the safety of both mother and fetus. Therefore, an objective, reliable computational mechanism capable of integrating maternal electronic health records into measurable predictive risk information is urgently required.

With the acceleration of artificial intelligence technologies, the utilization of machine learning methods has emerged as a state-of-the-art solution to transform clinical data into data-driven decision support models [1]. This data-driven approach has proven highly capable of efficiently identifying risk factors during early pregnancy, which in turn contributes significantly to reducing adverse perinatal outcomes [1]. Research regarding the application of classification algorithms for pregnancy risk management has been widely explored globally, including the development of large-scale preterm birth risk classification models using national population data [2]. Furthermore, machine learning-based risk stratification techniques have been successfully implemented to identify high-risk patients in other specific scenarios, such as the management of gestational diabetes complications and the prediction of macrosomia incidence using tree-based decision algorithms [3].The success of these literatures confirms that intelligent algorithms possess high capabilities in handling the complexity of maternal biological parameters.

In the predictive modeling of clinical pregnancy risks, the Random Forest and Logistic Regression algorithms are frequently deployed as the two most prominent comparative approaches favored by researchers [4]**.**For instance, in detecting the adverse outcomes of preeclampsia complications, the combination of logistic regression models and tree-based ensemble algorithms has proven capable of delivering high predictive accuracy to assist clinicians in identifying patients in critical conditions [4]. Nevertheless, a tangible challenge in applying machine learning models within local medical domains is the gap between raw algorithmic predictions and strict compliance with the established clinical guidelines enforced at local healthcare centers. Additionally, most prior studies still focus on passive experimental environments and have not yet been directed toward the deployment phase of models into digital web services accessible in real-time by healthcare professionals at primary clinics.

To address these research gaps, this study proposes a hybrid approach that integrates clinical expert systems (*maternal clinical rule-based labeling*) with the optimization of the Random Forest algorithm as the primary model and Logistic Regression as the baseline model. This clinical rule-based logic is systematically structured to encompass vital maternal parameters such as hemoglobin (Hb) levels, systolic and diastolic blood pressure, upper arm circumference (LiLA), body mass index (BMI), age factors, and comorbidity history to lock clinical validity onto the target labels. Utilizing a dataset consisting of 10,000 comprehensive ANC records, the class imbalance issue within the risk distribution is mitigated using a random oversampling technique before optimizing the models through GridSearchCV hyperparameter tuning based on a 5-fold cross-validation. The primary objective of this research is to evaluate the comparative accuracy of both models and integrate the highest-performing architecture into a RESTful web service framework via FastAPI and a Streamlit interactive interface under a system called "MamaCare AI". Experimental results demonstrate that the optimized Random Forest model yields a superior classification accuracy of 98.50%, successfully outperforming the Logistic Regression baseline at 95.65%. This contribution provides a responsive, applicable, and standard-compliant clinical decision support system to accelerate early pregnancy risk screening in the field.

2. Methods

The research workflow is structured systematically into several key blocks, beginning with data acquisition, followed by preprocessing, target engineering, feature engineering, class balance handling, and model optimization. The proposed hybrid framework integrates a maternal clinical expert system with tree-based machine learning architecture to achieve robust risk stratification. The overall experimental setup and data pipeline are described descriptively within the subsequent subheadings to ensure comprehensive reproducibility.

2.1 Dataset and Clinical Preprocessing

The primary dataset utilized in this study consists of 10,000 comprehensive maternal clinical records compiled from Antenatal Care (ANC) examinations. The raw data contains unstructured data blocks and missing clinical values that require specific preprocessing pipelines before feature extraction. The temporal features are adjusted by transforming raw birth dates into a normalized maternal age attribute, denoted as *Usia_Ibu*, which is clipped within a clinically viable range of 14 to 55 years to suppress extreme measurement anomalies. Missing data points within the uterine fundus height measurements (*Tinggi Fundus Uteri*) are addressed using a localized statistical approach. Instead of applying a global mean, the missing values are imputed using the median values calculated specifically per trimester group to maintain physiological relevance across different gestational periods.
Furthermore, missing categorical elements within the clinical action attribute are systematically imputed with a default baseline label to ensure structural completeness across all rows. After executing the initial preprocessing pipeline, the total dataset retains its structural integrity with zero remaining missing values across the 10,000 patient records. This clean data matrix serves as the foundation for the subsequent target engineering and feature expansion phases.

2.2 Target Engineering via Maternal Clinical Rule-Based Logic

The core innovation of the proposed method lies in the implementation of a deterministic target engineering block derived from authoritative maternal health guidelines. The target classification label, denoted as RISIKO, is dynamically generated through a custom clinical expert system function that maps heterogeneous vital signs into three distinct operational risk categories, namely Normal (Class 0), Need Treatment (Class 1), and Need Referral (Class 2). The labeling architecture enforces an absolute override logic for critical conditions, classifying a patient directly into Class 2 if they present a krisis hypertension characterized by a systolic blood pressure >159 mmHg or a diastolic blood pressure >109 mmHg. Absolute inclusion into Class 2 is also triggered by severe anemia where hemoglobin levels fall below 8.0 g/dL, reactive maternal triple elimination tests for HIV or Syphilis, or a history of severe systemic diseases including heart conditions, chronic hypertension, and tuberculosis.

If the critical conditions for an immediate medical referral are not met, the system transitions into an accumulative clinical scoring mechanism to determine the boundaries for Class 1. Preeclampsia indicators such as moderate hypertension add 3 points, while pre-hypertension and a maternal age risk factor under 19 or over 38 years add 1 point to the total clinical score. Mild anemia, mid-arm circumference showing a risk of chronic energy deficiency (LiLA <23.5 cm), and extreme body mass index anomalies contribute an additional 2 points to the evaluation matrix. Obstetric history variables, including a history of two or more abortions, grand multiparity where the pregnancy count reaches five or more, and an inadequate first ANC visit also increment the score by 1 point. The final rule-based decision block designates a patient as Class 1 if the cumulative scoring matrix aggregates to a value equal to or greater than 4, while records falling below this threshold are classified as Class 0.

2.3. Feature Engineering and Expanded Feature Set

To maximize the predictive capacity of the downstream machine learning models, the clean data matrix is subjected to extensive feature engineering to capture hidden non-linear physiological interactions. New continuous indicators are derived mathematically from raw clinical variables, including Mean Arterial Pressure (*MAP*) computed from systolic and diastolic inputs, and Pulse Pressure (*Pulse_Pressure*) which captures the physiological difference between the two pressure readings. The categorical trimester text is mapped into an explicit numerical feature matrix (*Trimester_Num*) to allow numerical processing. Continuous clinical thresholds are further mirrored into explicit binary flags to represent severe anemia, hypertension flags, and obstetric risk conditions such as grand multiparity and high-risk abortion history.

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Table 1. Engineered Feature Matrix Configuration

Category	variable	Data Type	Description
Demographics	Usia_Ibu, Usia Kehamilann, Trimester_Num	Numerical	Maternal age, gestational age and encoded trimester index
Obstetric History	Gravida,Para, Abortus,Abortus_Tinggi,Grandemulti	Numerical	Pregnancy history counts, aprity,abortion frequency and high-risk flags
Anthropometry	IMT, LiLA (cm), Tinggi Fundus Uteri (cm),IMT_Kurus,IMT_Obese, KEK_Flag, KatIMT_EncKatLiLA_Enc	Numerical	Body mass index, mid-arm circumference, fundal height, and nutritional risk status indicators.
Vital Signs	TD Sistolik, TD Diastolik, MAP, Pulse_Pressure, Hiper_Flag, Hiper_Berat, KatTD_Enc	Numerical	Blood pressure measurements, derived arterial pressures, and hypertension severity tags.

2.4 Class Imbalance Mitigation and Model Optimization Pipeline

Prior to the model training phase, the clean dataset of 10,000 records is partitioned using a stratified split strategy, allocating 80% of the samples for training operations and retaining the remaining 20% as an independent testing matrix. The initial training partition exhibits a severe class imbalance inherent to real-world maternal health data distributions, where normal cases heavily outnumber critical high-risk records. This imbalance presents a significant operational constraint that can degrade the sensitivity of traditional machine learning classifiers toward minority risk groups. To resolve this issue, an algorithmic random oversampling technique is applied strictly within the training preprocessing pipeline. The minority risk classes are resampled with replacement until their row counts precisely match the majority class frequency, establishing a perfectly balanced dataset of 24,000 samples across the three classes to ensure unbiased split calculations.

The optimization pipeline utilizes an exhaustive hyperparameter tuning block executed via GridSearchCV under a stratified 5-fold cross-validation setup to maximize the weighted F1-score. For the primary Random Forest architecture, the tuning grid comprehensively evaluates variations in estimator counts ranging from 50 to 200, maximum tree depths, split criteria using Gini impurity and entropy, and maximum feature sampling strategies. Simultaneously, the baseline Logistic Regression model is optimized by searching regularized cost parameters and inverse regularization strength across multiple iterations. Model validation is conducted rigorously across all folds to prevent overfitting, and the final classification capacity of the optimized estimators is evaluated against the independent test matrix using accuracy, precision, recall, and weighted F1 metrics.

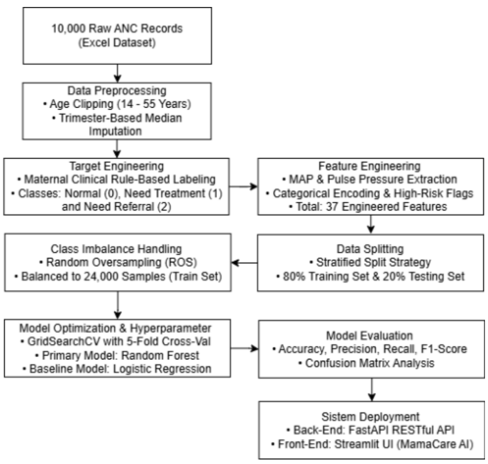

Figure 1. Proposed MamaCare AI Research Framework and Data Pipeline

2.5 System Deployment Architecture

To ensure the practical utility and scalability of the optimized machine learning models, a downstream production deployment phase is incorporated into the research framework [1]. The highest-performing model configuration is serialized into a reusable artifact and exposed as a micro-service through a high-performance RESTful Application Programming Interface (API) built using the FastAPI framework [3]. This backend architecture enforces asynchronous query handling to ensure rapid, low-latency predictive responses when handling simultaneous telemetry inputs from multiple primary healthcare endpoints. The API payload is strictly structured to receive raw JSON data parameters reflecting the maternal clinical variables, internally compute the derived features, and return the predicted clinical risk classification alongside the classification probabilities.

The frontend interactive user interface is developed utilizing the Streamlit framework, establishing a clean, responsive, and intuitive web application tailored for medical practitioners in rural clinics [1]. This interface abstracts the complex computational layers of the underlying tree-based ensemble, allowing healthcare workers to easily enter anthropometry data, vital signs, and laboratory diagnostics into structured form fields. Upon form submission, the web application dispatches a synchronous HTTP POST request to the FastAPI backend, retrieves the computed risk tier, and dynamically updates the visual dashboard to present clear, actionable medical alerts[3].Ultimately, transforming these analytical machine learning weights into usable, open-source micro-services establishes a scalable conceptual paradigm for modern maternal healthcare, successfully bridging the gap between raw experimental algorithms and compliant real-time diagnostic systems [5].

3. Results and Discussions

This section presents the empirical findings of the predictive models and critically interprets their clinical implications by contrasting the results against recent benchmarking literatures. The evaluation focuses on the classification capability of the optimized Random Forest model against the Logistic Regression baseline. To ensure a comprehensive discussion, the performance parameters are cross-examined with state-of-the-art maternal risk stratification systems developed across various geographic and resource-constrained clinical settings.

3.1 Experimental Result and Performance Analysis

The experimental evaluation on the independent test matrix demonstrates that the optimized Random Forest model delivers an exceptional classification capacity, achieving an overall accuracy score of 98.50%. This performance significantly outperforms the baseline Logistic Regression model, which yields a lower accuracy of 95.65%, proving that tree-based ensemble methods possess a superior ability to resolve complex non-linear boundaries within high-dimensional maternal physiological data streams [6].The integration of the deterministic target engineering block via maternal clinical rule-based labeling successfully locks clinical validity onto the classification process, preventing the model from generating pathologically counter-intuitive risk outputs. The fine-tuning phase via GridSearchCV with 5-fold cross-validation effectively stabilizes the decision tree branches, yielding a robust cross-validation score of 0.9879±0.0013 and maintaining high precision and recall balances across all three target classes, namely Normal, Need Treatment, and Need Referral.

The detailed distribution of the classification errors for the primary Random Forest model is visually represented through the confusion matrix illustrated in Figure 2. The diagonal elements of the matrix confirm that misclassified observations are minimized to a negligible fraction, validating the model's robustness against severe class imbalance constraints that typically plague medical datasets [7]. Feature importance analysis indicates that systolic and diastolic blood pressure, hemoglobin levels, and the cumulative engineered clinical risk indicator contribute the highest statistical weights during the node split calculations. This high information density proves that utilizing non-invasive vital parameters combined with laboratory diagnostics provides a reliable baseline for automated point-of-care screening. To provide a transparent empirical compilation of the experimental testing results, the final classification metrics, F1-scores, and cross-validation boundaries captured directly from the environment terminal are documented comprehensively in Figure 2.

```
=======================================================
 ◆ RANDOM FOREST PERFORMANCE (Setelah Tuning)
=======================================================
Akurasi  : 0.9850  (98.50%)
F1 Score : 0.9850
              precision    recall  f1-score   support

       NORMAL      0.98      0.97      0.98       674
PERLU TINDAKAN      0.98      0.98      0.98       807
 PERLU RUJUKAN      1.00      1.00      1.00       519

     accuracy                          0.98      2000
    macro avg      0.99      0.99      0.99      2000
 weighted avg      0.99      0.98      0.98      2000

Cross-Val (5-fold): 0.9879 ± 0.0013

=======================================================
 ◆ LOGISTIC REGRESSION PERFORMANCE (Setelah Tuning)
=======================================================
Akurasi  : 0.9565  (95.65%)
F1 Score : 0.9566
              precision    recall  f1-score   support

       NORMAL      0.93      0.95      0.94       674
PERLU TINDAKAN      0.95      0.94      0.95       807
 PERLU RUJUKAN      1.00      0.99      1.00       519

     accuracy                          0.96      2000
    macro avg      0.96      0.96      0.96      2000
 weighted avg      0.96      0.96      0.96      2000
```

Figure 2. Proposed MamaCare AI Research Framework and Data Pipeline

3.2 Discussion and Validation Againts Existing Literature

The exceptional accuracy of 98.50% achieved by the proposed Random Forest framework offers a substantial technical advancement compared to previous high-risk pregnancy prediction models [7].In low-resource settings, traditional decision tree architectures historically achieved a validation score of 98.8% using limited un-supplemented sets of clinical vitals [7]. The high capability of the proposed method is attributed to the extensive feature engineering phase that expanded the data matrix into 37 distinct parameters, capturing intricate physiological interactions such as Mean Arterial Pressure and trimester-specific fundal height trends. Furthermore, while demographic factors like residency or income show minimal direct statistical correlation with risk scores, maternal educational status remains a major determinant that is heavily associated with the prevalence of severe prenatal complications [8].This alignment confirms that risk screening instruments must incorporate holistic socio-demographic features alongside vital parameters to provide well-calibrated predictions within highly diverse socioeconomic subsets [9].In handling such complex clinical features, tree-based ensemble frameworks consistently prove their robustness by maintaining balanced precision-recall matrix weights and establishing sharp non-linear decision borders without overfitting the evaluation dataset [7].

The successful implementation of the multi-class stratification scheme also addresses critical gaps highlighted in recent maternal health systematic reviews [5]. Prior predictive modeling efforts have frequently been constrained by passive experimental environments, failing to bridge the operational gap between raw algorithmic training and practical clinical deployment [5]. By embedding the optimized tree-based weights into an open-source digital ecosystem powered by a FastAPI backend and a Streamlit interactive dashboard, this research directly materializes a usable Clinical Decision Support System (CDSS). This deployment architecture effectively mitigates the risk of diagnostic delays and manual assessment subjectivity, offering frontline healthcare workers in rural clinics a responsive, standard-compliant screening tool to optimize medical referrals and safeguard maternal-neonatal outcomes [8], [5].

4. Conclusions

This research has successfully developed and evaluated "MamaCare AI", a standard-compliant and robust hybrid clinical decision support system designed to accelerate early pregnancy risk screening in primary healthcare facilities. By integrating a deterministic maternal clinical rule-based expert system for target engineering, the structural validity of the clinical parameters was successfully embedded into the data logic, effectively neutralizing the risk of pathologically counter-intuitive risk outputs. The experimental results demonstrate that the optimized Random Forest model delivered an outstanding classification performance, achieving an overall accuracy score of 98.50%. This tree-based ensemble approach significantly outperformed the baseline Logistic Regression model, which yielded a lower accuracy of 95.65%, thereby confirming the superior capability of tree-based architectures in resolving complex, non-linear physiological boundaries within maternal electronic health records without suffering from overfitting.

Beyond its high computational validation scores, this study successfully bridged the operational gap identified in prior passive experimental literatures by deploying the optimized machine learning weights into a live digital web ecosystem. Powered by a high-performance, asynchronous FastAPI backend and a responsive Streamlit interactive user interface, the developed system materializes a concrete, applicable tool for front-line healthcare workers in resource-constrained rural clinics. This deployment framework successfully eliminates manual subjectivity and significantly reduces the likelihood of diagnostic delays, providing medical professionals with a reliable and standard-compliant mechanism to optimize medical referrals. For future research directions, the predictive architecture can be further enhanced by incorporating unstructured health data and validating the longitudinal model calibration across more diverse geographical and regional patient demographics.

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